

# February 2026 EDA Summary

## NYC TLC High-Volume For-Hire Vehicle (HVFHV) Trip Records

### 1. Dataset Overview

The February 2026 dataset contained **19,875,686 trips across 28 days**, using the same 25-column TLC HVFHV structure as January. The two providers were **HV0003 (14.17M trips)** and **HV0005 (5.71M trips)**.

No records were removed during February processing. The final analytical dataset contained **19,875,686 rows and 36 columns**. Data-quality issues such as negative financial values, timestamp inconsistencies and zero-time trips were investigated and retained with appropriate flags rather than being aggressively deleted.

A notable timestamp finding was that **17.97% of records did not exactly match timestamp-derived trip duration**, but most differences were very small. Only **594 records differed by more than 5 minutes**, indicating that the large mismatch percentage was largely driven by timestamp precision rather than necessarily invalid trips. The pickup-to-dropoff sequence remained valid, with **zero trips having drop-off before pickup**.

---

### 2. Key EDA Findings

#### Demand Patterns

February showed a strong and recurring time-of-day demand pattern.

- **6 PM:** highest demand — **1,196,467 trips**
- **3 AM:** lowest demand — **265,425 trips**

Demand was relatively low overnight, increased throughout the morning and afternoon, and peaked around the evening period.

Trip characteristics varied considerably by hour. While **6 PM had the highest trip volume**, average trip duration peaked earlier at **3 PM (22.29 minutes)**. Average trip distance was highest at **5 AM (6.82 miles)** and lowest at **6 PM (4.08 miles)**.

This reinforces an important analytical distinction: **peak demand does not necessarily mean peak trip duration or distance**.

#### Overall Trip Economics

| Metric             | February         |
|--------------------|------------------|
| Total Trips        | <b>19.88M</b>    |
| Total Trip Miles   | <b>92.41M</b>    |
| Average Distance   | <b>4.65 mi</b>   |
| Average Duration   | <b>19.38 min</b> |
| Total Base Fare    | <b>\$529.30M</b> |
| Average Base Fare  | <b>\$26.63</b>   |
| Total Driver Pay   | <b>\$408.15M</b> |
| Average Driver Pay | <b>\$20.53</b>   |

The distance and duration distributions were strongly right-skewed. The **median trip distance was 2.69 miles** compared with a 4.65-mile mean, while the median duration was **15.82 minutes** compared with a 19.38-minute mean. This shows that a relatively small number of long trips pull the averages upward.

---

### 3. Provider & Geographic Analysis

#### Provider Performance

| Metric | HV0003 | HV0005 |
|--------|--------|--------|
|--------|--------|--------|

|                |                |         |
|----------------|----------------|---------|
| Trips          | 14.17M         | 5.71M   |
| Market Share   | <b>71.28%</b>  | 28.72%  |
| Avg Base Fare  | <b>\$27.51</b> | \$24.46 |
| Avg Driver Pay | <b>\$21.13</b> | \$19.07 |
| Avg Distance   | <b>4.83 mi</b> | 4.21 mi |

HV0003 handled approximately **7 out of every 10 trips** and recorded higher average fare, driver pay and trip distance than HV0005. Its share of total base fare was approximately **73.6%**, slightly above its 71.3% share of trips.

## Geographic Concentration

Demand remained concentrated in a relatively small number of pickup locations.

The top pickup locations included:

- **138:** 341,205 trips
- **132:** 273,543
- **61:** 259,773
- **76:** 241,178
- **79:** 229,729

The strongest geographic finding was on the **drop-off side**: Location **265 recorded 821,845 trips**, substantially more than the next destinations, including Location 138 with 365,192 trips.

This indicates a notable concentration of destination demand that can be investigated further once the TLC Location IDs are mapped to their corresponding NYC zones.

---

## 4. Trip Economics & Customer Behavior

### Distance and Fare

February showed a strong positive relationship between trip distance and average fare:

| <b>Trip Distance</b> | <b>Avg Base Fare</b> |
|----------------------|----------------------|
| 0–2 mi               | \$12.90              |
| 2–5 mi               | \$22.52              |
| 5–10 mi              | \$35.72              |
| 10–15 mi             | \$52.86              |
| 15–20 mi             | \$76.16              |
| 20+ mi               | <b>\$116.74</b>      |

Trips of 20+ miles had an average fare roughly **9×** that of trips under 2 miles. This makes trip distance an important dimension for understanding fare economics.

## **Tipping**

Average tips also increased with base fare:

| <b>Base Fare</b> | <b>Avg Tip</b> |
|------------------|----------------|
| Below \$10       | \$0.35         |
| \$10–20          | \$0.53         |

|          |               |
|----------|---------------|
| \$20–40  | \$1.07        |
| \$40–60  | \$2.27        |
| \$60–100 | \$4.13        |
| \$100+   | <b>\$8.61</b> |

Higher-fare trips were associated with higher average tips. This is a descriptive relationship and does not establish that higher fares cause higher tips.

Shared rides remained a relatively small component of demand, with **385,765 shared requests**, representing approximately **1.94% of February trips**.

---

## 5. February Business Takeaway

February 2026 continued to show **strong evening demand, concentrated geography and clear differences in provider-level trip economics**. HV0003 handled the majority of trips and also recorded higher average fare, driver pay and distance.

Most trips were relatively short, but **longer trips were associated with substantially higher fares**, while higher-fare trips also showed higher average tips. Geographic demand was concentrated across recurring pickup locations, with **Location 265 standing out as a particularly dominant drop-off destination**.

The February analysis also reinforced several themes already observed in January:

**Demand by Time → Provider Performance → Geographic Concentration → Trip Economics → Customer Behavior**

These recurring patterns were carried forward for comparison with March before finalizing the Q1 SQL analysis.